

# MFBRU-Net: Multi-scale Feature Fusion and Boundary Refining U-Net for Medical Image Segmentation<sup>★</sup>

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## ABSTRACT

Medical image segmentation, as a core task in intelligent diagnostic systems, imposes stringent demands on models' capabilities in spatial structure modeling, semantic consistency preservation, and detail reconstruction. Although deep learning-based methods have achieved progress on standard datasets, prevalent challenges in public datasets—such as blurred structural boundaries, densely distributed small objects, and highly diverse target scales—still lead to deficiencies in existing approaches regarding multi-scale semantic fusion, long-range context modeling, and fine boundary perception. To address these issues, this paper proposes MFBRU-Net, an end-to-end segmentation framework incorporating three core innovations: a Multi-scale Feature Fusion Module (MFFM) for cross-level geometric-semantic collaboration; a Saliency-guided Feature Refinement module (SFR) to enhance sensitivity to noise and local saliency; and an xLSTM Boundary Refinement Decoder (xBRD) for improved fine-grained boundary recovery. Experiments on five representative datasets—Kvasir-SEG, CVC-ClinicDB, ETIS, GlaS, and BUSI—demonstrate that MFBRU-Net achieves high segmentation accuracy across various scenarios, with Dice scores of 93.73%, 95.26%, 95.50%, 94.30%, and 83.21%, respectively, and a HD95 of only 0.10 on ETIS, significantly outperforming most comparative models. Moreover, it improves average Dice by 1.8% (range: 1.2%–3.5%) and IoU by 2.1% over mainstream methods. In conclusion, MFBRU-Net offers an efficient solution for segmenting complex medical images—including endoscopic, pathological, and ultrasound data—and exhibits notable performance in boundary reconstruction and small object segmentation.

## 1. Introduction

In recent years, the rapid advancement of artificial intelligence has driven significant breakthroughs in medical image analysis through deep learning. Particularly in medical image segmentation [1, 2, 3, 4, 5, 6], this technology plays a vital role in critical clinical applications such as tumor detection, organ recognition, and lesion localization, establishing itself as a core component of intelligent diagnostic systems. However, despite the strong performance of U-Net [1] and its derivative architectures on publicly available datasets, they still exhibit three common limitations in complex medical scenarios.

First, in medical image segmentation tasks, the information contained in feature maps of different scales varies significantly: Shallow features possess rich edge and texture details but have limited semantic expression capabilities; In contrast, deep features contain abstract global semantics but have low spatial resolution, making it difficult to capture fine-grained structures. Although existing methods, such as U-Net and DUCK-Net [7], generally fuse multi-scale features [8] through skip connections, feature concatenation, or simple weighting, these static operations essentially fail to address the issues of semantic misalignment and structural

mismatch between cross-layer features. This easily leads to blurred feature expression after fusion and even generates redundant or mutually interfering semantic information. Moreover, since the distribution structures of features at different scales in the channel dimension are often inconsistent, direct concatenation tends to cause channel redundancy or semantic dilution, limiting the further improvement of the network's semantic discriminative ability.

Second, effective semantic consistency modeling not only requires features to have global semantic perception capabilities during the encoding process, but also demands unified regulation and feedback of such semantic information in the decoding and reconstruction stages, forming a top-down semantic transmission loop. However, most current methods limit semantic modeling to local attention or shallow supervision, lacking a continuous and consistent guidance mechanism along the encoding-to-decoding path. This leads to issues of drift, redundancy, and omission in fused features. Such semantic inconsistency is particularly prominent when multiple types of targets coexist or lesion boundaries are unclear, ultimately resulting in the model's insufficient discriminative power for key regions and target recognition being easily disturbed by context. Although some works attempt to construct long-range dependencies through global channel attention [9, 10, 11] or Transformer mechanisms [12, 13, 14, 15], their structural complexity is high, and they lack adaptation strategies for multi-scale semantic differences, making it difficult to achieve collaborative modeling on a unified path.

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Finally, medical images often have problems such as complex tissue and organ structures, blurred boundaries[16], and large variations in target scales. Although the step-by-step upsampling and convolution operations in traditional decoding paths can restore spatial resolution to a certain extent, in the absence of sufficient semantic supervision or context guidance, they easily lead to blurred or discontinuous target edges[17], seriously affecting the segmentation accuracy in structure-sensitive tasks (such as small targets like polyps[18, 19] and tumors). Especially in the decoding process of deep features, since the original details have been repeatedly abstracted and compressed, existing methods that rely solely on upsampling interpolation are difficult to restore the real contours of targets. In addition, the decoder does not fully model the semantics of upper and lower layers, resulting in structural drift or semantic degradation of information flow during backpropagation. Some models try to introduce attention mechanisms[20] or boundary guidance, but most are limited to explicit guidance rather than structural modeling, failing to provide support in terms of temporal modeling or context memory. Therefore, there is an urgent need to introduce structural modules with dynamic modeling capabilities to enhance the adaptability of the decoding process to context changes and boundary details[21], and to achieve continuous structure restoration in regions with complex morphologies.

To address the aforementioned issues, this paper proposes a novel method based on a systematic review of current research progress and bottlenecks. While retaining the U-shaped backbone structure, this method introduces three key modules, respectively targeting semantic perception enhancement in the encoding stage, multi-scale feature-guided fusion in the intermediate stage, and temporal context modeling in the decoding stage. Through the collaborative design of these modules, the model can not only extract spatial details from shallow structures but also achieve high-level modeling on deep semantics. Furthermore, it completes high-quality restoration of boundary structures in the upsampling stage, comprehensively improving the accuracy of the model in medical image segmentation. Meanwhile, it has achieved leading segmentation performance on multiple public medical image datasets.

In summary, aiming at the deficiencies of existing medical image segmentation models in multi-scale semantic fusion, context consistency modeling, and structural detail restoration, this paper proposes the following four core improvements:

- **MFBRU-Net** is developed as an enhanced segmentation framework that builds upon a three-level cascaded U-Net backbone, incorporating deformable convolution and gating mechanisms to achieve a balance between segmentation accuracy and computational efficiency.
- The **Multi-scale Feature Fusion Module (MFFM)** performs cross-stage feature aggregation combined

with geometric–semantic calibration, enabling consistent multi-resolution integration and adaptive channel distribution adjustment, which reduces the misalignment between low-level details and high-level semantics.

- The **Saliency-Guided Feature Refinement Module (SFR)** integrates an EMA-based multi-scale attention mechanism with residual enhancement units to emphasize salient spatial–channel regions, while four-layer convolutional extraction further improves cross-level feature discrimination.
- The **xLSTM Boundary Refinement Decoder (xBRD)** introduces a bidirectional state propagation strategy. By reformulating feature maps into spatial sequences, it applies exponential gating (input gate for absorbing new evidence, forget gate for filtering outdated information) to refine boundary topology more effectively.

## 2. Related Work

### 2.1. Analysis of CNN-based Feature Processing Mechanisms

Convolutional Neural Networks (CNNs)[1, 2, 3, 22, 23] have been widely used in medical image segmentation due to their excellent local perception and spatial modeling capabilities. Among them, U-Net, based on an encoder-decoder structure, fuses shallow edge information[24, 25, 25, 26, 27] with deep semantic features through skip connections, showing robust performance on small-sample medical datasets. However, its multi-scale fusion mechanism[28, 29] mainly relies on static stacking and feature concatenation, which struggles to effectively capture semantic differences between layers, leading to redundant information accumulation and structural mismatch.

To alleviate these limitations, U-Net++[22] introduces dense connections, constructing multi-level residual paths between different scales to enhance the flow of semantic information and improve the ability of shallow features to support high-level semantics. Nevertheless, it essentially remains a static connection without an explicit alignment mechanism, resulting in redundancy and drift in fused features. PraNet[4], targeting small-object and boundary-sensitive tasks, adopts a reverse erasure strategy to guide the model to focus from the background to the target region, demonstrating certain advantages in improving segmentation accuracy. However, its multi-scale path[30, 31, 32, 33] is singular, lacking an adaptive adjustment mechanism between semantics. DUCK-Net[7] models details and semantics separately through a dual-branch architecture, enhancing structural modeling capabilities. Yet, it fails to explicitly address scale alignment between layers, and its non-local mechanism incurs high computational costs, making it susceptible to data scale limitations. KM-UNet[6] introduces multi-level attention mechanisms and resolution-guided paths on the basis of U-Net, strengthening the collaborative modeling of semantic consistency and

boundary details, and is suitable for medical images with complex structures. However, its feature fusion remains a static structure, lacking dynamic adaptability to contextual semantics. PARF-Net[5] enhances semantic supplementation and redundancy suppression capabilities through multi-scale parallel receptive fields, exhibiting good adaptability in complex deformation scenarios. However, its fusion process focuses more on the channel level, with limited spatial alignment capabilities, posing a risk of semantic mismatch.

In summary, current CNN-based segmentation models have explored various approaches in multi-scale modeling, ranging from connection enhancement (U-Net++[22]), attention guidance (PraNet[4]), dual-branch structures (DUCK-Net) to multi-level structure integration (KM-UNet[6], PARF-Net[5]). These efforts have driven fusion strategies to evolve from static concatenation to structure-guided methods. However, most methods still rely on preset paths and fixed structures[28, 29], lacking adaptive recognition of semantic differences across scales and spatial alignment mechanisms, which hinders their ability to fully express the boundaries, morphology, and continuity of target regions.

Therefore, the proposed Multi-scale Feature Alignment and Fusion Module (MFFM) aims to construct a novel fusion mechanism with hierarchical perception, semantic guidance, and dynamic regulation capabilities. It effectively aligns semantic levels between multi-scale features, suppresses redundant information expression, and thereby improves the segmentation accuracy and generalization ability of the model in complex medical image scenarios.

## 2.2. Analysis of Transformer-based Feature Processing Mechanisms

In recent years, the Transformer[34, 35, 36] architecture has been widely applied in medical image segmentation due to its superior global modeling capability[37], demonstrating significant advantages in long-range dependency modeling and structural perception. However, in high-precision segmentation tasks, existing Transformer models still face issues such as insufficient semantic consistency modeling, limited cross-scale semantic integration, and poor adaptability to spatially heterogeneous structures—particularly evident in the decoding stage.

Early models represented by TransUNet[13], although introducing Vision Transformer into the encoder to effectively enhance global perception, still rely on static concatenation for their fusion strategy, lacking a semantic selection mechanism. This leads to inadequate dynamic regulation of responses between multi-scale semantics, discontinuous cross-layer information transmission, difficulty in restoring target boundary details, and a tendency to miss small targets or cause semantic drift. DA-TransUNet[38] incorporates dual attention mechanisms to strengthen the modeling of key regions, but the attention path is focused on middle-layer modulation, lacking collaborative regulation of full-process semantics between encoding and decoding paths, making it difficult to achieve effective fusion of shallow and deep information.

Trans2UNet[39] attempts to enhance semantic guidance through a dual-branch structure and multi-scale convolution enhancement modules, but its fusion method still relies on static stacking, struggling to adapt to discontinuities in expression caused by scale variations and complex structures. FCBFormer[14] and UniNet[40] introduce spatial coordinate awareness and dynamic weighting mechanisms, alleviating modal differences in local structures to a certain extent. However, their semantic modeling tends to focus on spatial position perception, lacking the ability to connect information across semantic levels, resulting in fragmented feature expression when processing targets with drastic morphological changes or irregular boundaries. Topo-VM-UNetV2[15] introduces topology-guided paths and position preservation mechanisms to model morphological continuity between targets, but its information transmission is unidirectional, lacking semantic loop regulation, leading to structural fragmentation and semantic loss during cross-layer alignment.

In summary, while current Transformer-based[13, 41] segmentation models perform well in global perception and spatial modeling, they generally lack a closed-loop semantic regulation mechanism in the decoding stage. This results in problems such as structural fragmentation during semantic transmission, insufficient scale adaptation, and unstable recognition of complex targets. The root cause lies in the continued reliance on static stacking and local modulation in fusion strategies, which fail to support dynamic interaction and continuous modeling between different semantic levels.

To address this, the proposed MFBRU-Net introduces the Saliency-Guided Feature Refinement Module (SFR), which realizes progressive guidance of cross-layer semantic information and maintains semantic consistency in the decoding stage through semantic entropy-based dynamic attention paths[10, 42, 43]. The SFR module breaks away from traditional static concatenation by enhancing responses in small targets and low-semantic regions through saliency weight regulation, fundamentally alleviating the difficulty in recognizing weak-semantic targets. Additionally, the model incorporates a multi-scale semantic reconstruction structure[44] and residual semantic calibration mechanism in the decoding stage, strengthening interactive compensation between shallow spatial details and deep semantics. This achieves continuity of semantic flow, integrity of structural restoration, and adaptive unification of scale features, effectively resolving the decoding bottlenecks in existing Transformer architectures.

## 2.3. Application of xLSTM in Medical Image Segmentation

In medical image segmentation tasks, images such as breast ultrasound and colonoscopy often exhibit problems like blurred boundaries and complex lesion morphologies. This causes traditional convolutional structures to easily lose local details when extracting deep features, and especially insufficient modeling of edge regions. Although the U-Net[1] structure partially alleviates the loss of semantic information

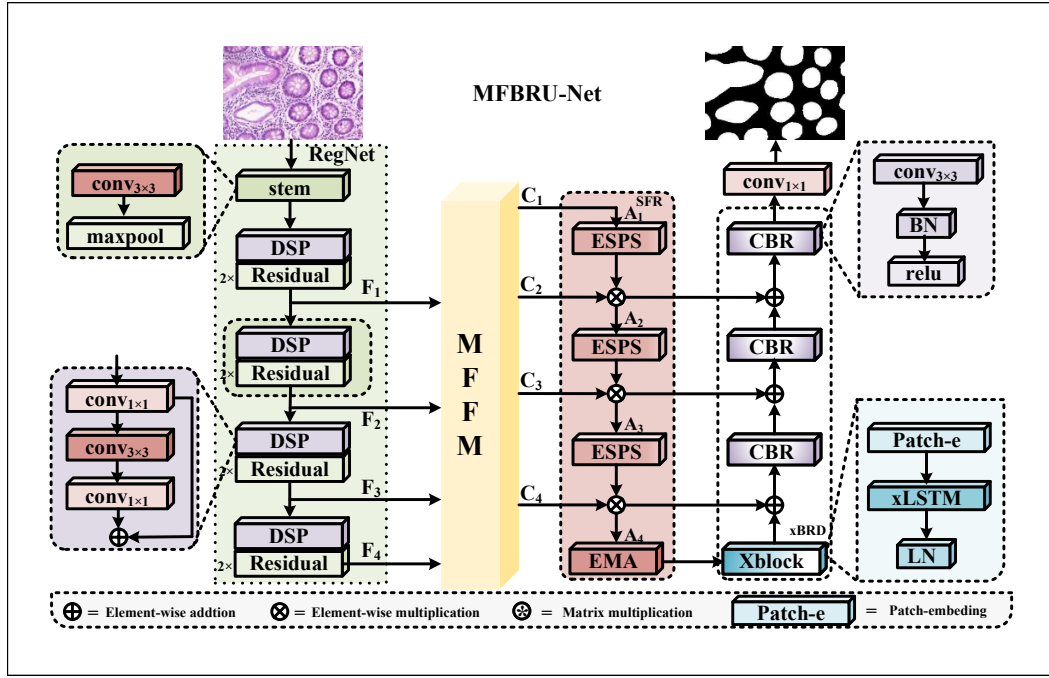


Figure 1: Overview of the MFBRU-Net architecture, illustrating the key modules and data flow.

through skip connections, its receptive field is limited, making it difficult to capture long-range dependencies between different regions in the image and restricting the model's performance in complex structures.

To enhance the model's spatial perception capability, this paper introduces and improves the xLSTM (eXtended Long Short-Term Memory)[45] structure, which was originally designed for natural language processing. xLSTM[45] features an enhanced long-term memory mechanism, enabling better modeling of nonstationary sequence information. Existing studies have embedded it into the encoder of U-Net to strengthen context expression, but issues remain, such as fixed modeling paths and a lack of multiscale fusion and regional adaptability.

To address these shortcomings, this paper transfers the "temporal modeling" capability of xLSTM[46, 47] to "spatial modeling" scenarios, proposing a context-modeling method based on spatial sequence construction. Specifically, fused multi-scale features are encoded into spatial pseudosequences, and dynamic information updates and state tracking are performed through the xLSTM module[48], thereby achieving semantic collaboration between remote regions. This mechanism effectively captures dependencies between structures such as lesions, edges, and backgrounds, improving the model's adaptability to heterogeneous regions and its ability to preserve details.

Compared with traditional attention mechanisms, xLSTM introduces state update and gating mechanisms, enabling better performance in maintaining long-term information and resisting semantic degradation. Meanwhile, this

structure naturally supports multiscale feature fusion, helping to maintain semantic consistency across different resolutions. The introduction of the module not only enhances the model's robustness and boundary expression capability, but also provides a new modeling path for segmentation tasks involving detailed and structurally variable medical images.

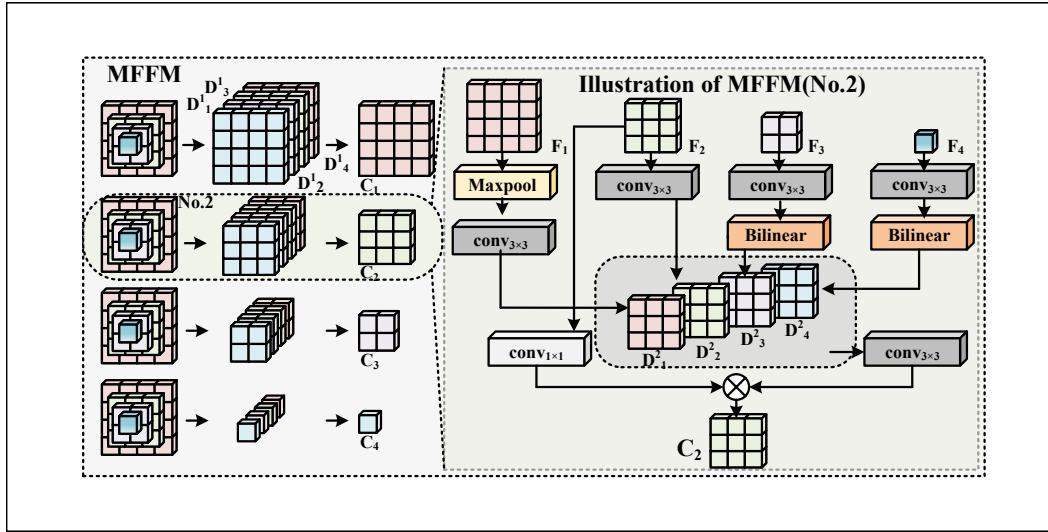
In summary, this paper innovatively extends xLSTM from the time domain to spatial modeling tasks, endowing the model with stronger context understanding and dynamic modeling capabilities. This significantly improves segmentation performance in complex medical image scenarios and represents one of the key breakthroughs in the structural design of MFBRU-Net.

### 3. Method

#### 3.1. Overall Model: MFBRU-Net

To address issues such as multi-scale semantic misalignment, spatial structure distortion, and poor boundary continuity, this paper designs an MFBRU-Net structure with deep collaboration between modules. A detailed schematic diagram of the model is shown in Figure 1. First, the encoder adopts an efficient RegNet backbone network to encode the input image into four levels of feature maps ( $F_1-F_4$ ), where shallow features retain rich spatial details and deep features carry abstract semantic information.

Aiming at the geometric misalignment and semantic inconsistency in multi-scale features, we propose the Multi-scale Feature Fusion Module (MFFM). This module performs spatial alignment of features at different levels based on a deformable convolution mechanism, while introducing



**Figure 2: MFFM Diagram:** Overview of the MFFM architecture, illustrating the key feature modules and data flow for understanding the overall structure.

channel distribution constraints to mitigate semantic deviation and improve the quality of feature fusion.

Furthermore, to enhance the model's ability to perceive small targets or regions with complex structures, the Saliency-Guided Feature Refinement Module (SFR) is introduced. This module combines multi-scale attention guidance and hierarchical convolution extraction to strengthen the model's recognition of key structures and improve the effect of cross-scale feature integration.

In the decoding stage, the xLSTM Boundary Refinement Decoder (xBRD) is designed. It converts high-dimensional features into spatial scanning sequences and uses an exponential gating mechanism to model boundary continuity, effectively restoring tissue contours with complex morphologies. Finally, the CBR convolution block further refines the mask edges, and high-resolution segmentation results are output through bilinear interpolation combined with a Softmax classifier.

The overall network constructs an inter-module cascaded correction mechanism through the information transmission path of "alignment—focus—topological reconstruction," realizing end-to-end adaptation throughout the process from encoder feature extraction, multi-scale fusion to mask generation. This fully meets the multiple requirements for fine structures, boundary continuity, and global semantic understanding in medical image segmentation.

### 3.2. RegNet as the Encoder

In this study, the encoder adopts the RegNet architecture, which features a regular structure and excellent computational efficiency. By automatically optimizing the network's width, depth, and group convolution parameters within a regularized design space, RegNet balances expressive capability and inference efficiency, demonstrating strong image representation ability. The input to the model is first processed through the Stem layer, then divided into four stages

(Stage 1 to Stage 4) in the RegNet encoding path, outputting four groups of feature maps with different semantic levels. Their spatial resolutions are 1/4, 1/8, 1/16, and 1/32 of the original image, respectively, while the number of channels gradually increases, constructing a multi-scale feature representation with clear hierarchical structure. To improve parameter utilization and enhance semantic consistency between features of different stages, the model reuses the same set of network parameters (i.e., shared weights) across the four stages. This design means that the network uses identical convolution kernels and operation modes when processing features of different scales, thereby maintaining the coherence of cross-stage semantic expression without additional parameter overhead, and effectively reducing the model size and computational burden. These multi-scale features are then fed as basic inputs into the cross-stage feature alignment module and fusion module, supporting subsequent context modeling and mask reconstruction tasks. The introduction of the RegNet encoder significantly improves the quality of feature extraction and the model's generalization ability, enabling the network to perform more excellently in tasks such as medical image segmentation that have high requirements for accuracy and robustness.

### 3.3. Multi-scale Feature Fusion Module (MFFM)

In medical image segmentation tasks, the full fusion of multi-scale semantic information is crucial for improving the model's ability to recognize structures and resolve boundaries. Due to significant differences in spatial resolution, number of channels, and semantic abstraction levels between features of different layers, direct fusion often leads to information mismatch or semantic interference. To address this, this paper designs a Multi-scale Feature Fusion

Module (MFFM), which effectively alleviates structural inconsistencies between multi-layer features by aligning spatial scales and unifying channels of shallow, middle, and deep features. The module first maps features from different stages to a unified scale through convolution, pooling, and upsampling operations, then concatenates and fuses them in the channel dimension, ultimately achieving semantic alignment and information complementarity of features. The MFFM enhances cross-layer semantic flow and improves the overall segmentation network's comprehensive performance in recognizing local fine structures and modeling global contexts. The specific workflow is illustrated in Figure 2.

#### Implementation Process of MFFM

The four feature maps  $F_1$ – $F_4$  extracted by the backbone network RegNet are processed by MFFM to obtain features  $C_1$ – $C_4$ . Their spatial dimensions are consistent with the corresponding input features  $F_1$ – $F_4$ , maintaining 1/4, 1/8, 1/16, and 1/32 of the original input, respectively.

As illustrated in Figure 2, the working process of MFFM is exemplified by fusing  $F_1$ – $F_4$  to obtain  $C_2$ . First, the MFFM module performs unified processing on the spatial and channel dimensions of the four-layer feature maps extracted from different stages of the RegNet, ensuring structural alignment of multi-scale features before fusion. For the shallow feature  $F_1$ , maximum pooling combined with  $\text{conv}_{3 \times 3}$  is used to compress the spatial dimension to match that of  $F_2$ , resulting in  $D_1^2 \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times 32}$ . This operation not only reduces the computational burden but also preserves detailed information, serving as a key step in establishing spatial consistency. For the middle-layer feature  $F_2$ , a  $\text{conv}_{3 \times 3}$  is used to expand the number of channels to 64 while keeping the spatial scale unchanged, forming  $D_2^2 \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times 64}$ .

$$D_1^2 = \text{conv}_{3 \times 3}(\text{MaxPool}(F_1)), D_2^2 = \text{conv}_{3 \times 3}(F_2) \quad (1)$$

The deep semantic feature  $F_3$  has a relatively smaller spatial dimension, so after convolution extraction, it is upsampled once to restore to the target spatial scale  $D_3^2 \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times 32}$ ; the deepest feature  $F_4$  contains the most abstract semantic information, which requires two consecutive upsampling operations followed by convolution to obtain  $D_4^2 \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times 16}$ .

$$\begin{aligned} D_3^2 &= \text{UP}(\text{conv}_{3 \times 3}(F_3)), \\ D_4^2 &= \text{UP}(\text{UP}(\text{conv}_{3 \times 3}(F_4))). \end{aligned} \quad (2)$$

At this point, the four scale-unified feature maps are aligned in the spatial dimension, and their channel dimensions are controlled within a moderate range, facilitating subsequent fusion. Next, the above four groups of features are concatenated in the channel dimension to form the initial fused feature  $C_{in}$ .

$$C_{in} = \text{Concat}(D_1^2, D_2^2, D_3^2, D_4^2) \quad (3)$$

The fused feature  $C_{in}$  is processed by  $\text{conv}_{3 \times 3}$  and then element-wise multiplied with  $F_2$  that has undergone  $\text{conv}_{1 \times 1}$  to obtain  $C_2$ , marking the end of the MFFM process.

$$C_2 = \text{conv}_{3 \times 3}(C_{in}) \otimes \text{conv}_{1 \times 1}(F_2) \quad (4)$$

### 3.4. Saliency-Guided Feature Refinement Module (SFR)

Multi-scale feature fusion is a core link in improving the performance of medical image segmentation. Traditional methods mostly adopt simple concatenation or weighted summation, ignoring differences in semantic depth and spatial structure between features of different layers, which leads to information redundancy and semantic drift, restricting the model's discriminative power and generalization ability. To address this, this paper designs the Saliency-Guided Feature Refinement Module (SFR), which optimizes cross-scale fusion from two dimensions—semantic guidance and dynamic response—through semantic distillation and multi-scale attention modulation.

The specific process is as follows: Taking the deep high-semantic feature  $C_1$  as the initial input  $A_1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 64 \times B}$ , it is processed by the ESPS module to obtain feature  $B_1$ . Subsequently,  $B_1$  is element-wise multiplied with the next-layer feature to generate the fused feature.  $A_2$  is further input into ESPS to generate  $B_2$ , which is then multiplied with the shallower feature  $C_3$  to produce  $A_3$ . By analogy,  $A_3$  is processed by ESPS and fused with the shallowest feature  $C_4$  to output  $A_4 \in \mathbb{R}^{\frac{H}{32} \times \frac{W}{32} \times 64 \times B}$ .  $A_4$  is then passed through a separate EMA to generate  $B_5$ , which is fed into the xBlock.

The ESPS includes Efficient Multi-scale Attention (EMA), semantic distillation block (simDB), Adaptive Average Pooling (AAP), and Sigmoid activation function, which are used to extract spatial saliency weights. This top-down semantic guidance mechanism realizes effective guidance and weighted fusion of high-level semantics on low-level details by transmitting spatial saliency weights layer by layer, significantly alleviating semantic drift and information conflict. The EMA module dynamically adjusts feature responses in spatial and channel dimensions, enhancing the detailed representation of target regions and suppressing noise. Finally, the SFR module outputs multi-scale features with both high semantic consistency and detailed spatial structure, providing the decoder with feature support that is highly discriminative and structurally coherent, thus significantly improving segmentation performance and generalization ability. The detailed process of the ESPS module is illustrated in Figure 3.

#### EMA Module

The EMA (Efficient Multi-scale Attention) module achieves joint enhancement of feature maps in both spatial and channel dimensions by combining group channel processing, spatial-guided attention, and local context modeling. This module effectively captures multi-scale contextual information and improves the consistency of representation through

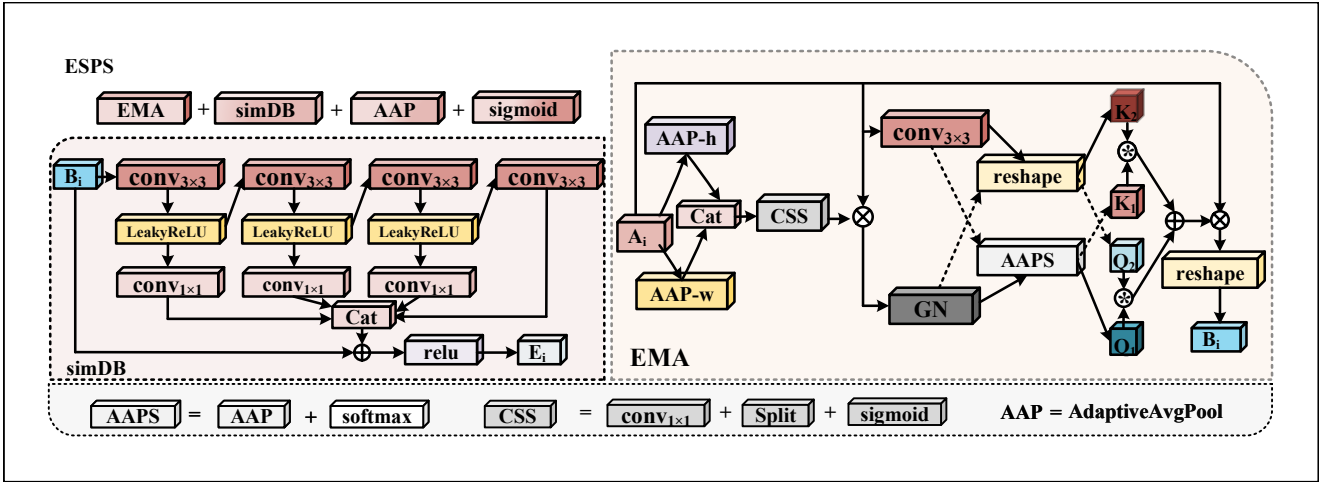


Figure 3: Schematic of the ESPS

group normalization, thereby enhancing the discriminative ability of features.

To describe its process in detail, we use the transformation from  $A_1$  to  $B_1$  to illustrate its working mechanism. The core idea is to reduce the number of channels of the input feature tensor  $A_1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 64 \times B}$  through group convolution, obtaining  $A_g \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times \frac{C}{G} \times (B \times 8)}$  (where  $C = 64$  and  $G = 8$ ) to improve computational efficiency. Subsequently, for the spatial dimension, the EMA module performs adaptive average pooling on  $A_g$  along the height and width directions, respectively, to obtain  $A_h \in \mathbb{R}^{\frac{H}{4} \times 1 \times \frac{C}{G} \times (B \times 8)}$  and  $A_w \in \mathbb{R}^{1 \times \frac{W}{4} \times \frac{C}{G} \times (B \times 8)}$ .

$$A_h = AAP - h(A_g), A_w = AAP - w(A_g) \quad (5)$$

After concatenating the two in the spatial dimension, they are mapped through  $conv_{1 \times 1}$  to obtain  $H_w \in \mathbb{R}^{(\frac{H}{4} + \frac{W}{4}) \times \frac{C}{G} \times (B \times 8)}$ .

$$H_w = conv_{1 \times 1}(Concat(A_h, A_w)) \quad (6)$$

The result is split into  $H \in \mathbb{R}^{\frac{H}{4} \times \frac{C}{G} \times (B \times 8)}$  and  $W \in \mathbb{R}^{\frac{W}{4} \times \frac{C}{G} \times (B \times 8)}$ , representing spatial weights in the height and width directions, respectively. Non-linear transformation is applied using Group Normalization (GN) and the Sigmoid activation function to generate  $X_1$ . Meanwhile,  $A_g$  is processed through  $conv_{3 \times 3}$  to obtain  $X_2$ .

$$X_1 = GN(A_g \otimes \sigma(H) \otimes \sigma(W)), X_2 = conv_{3 \times 3}(A_g) \quad (7)$$

Finally, to enhance local detail information, the input feature generates an additional feature map through  $conv_{3 \times 3}$ . It then calculates channel attention weights by combining this map with the result of matrix multiplication and summation of  $Q_1, Q_2, K_1$ , and  $K_2$ —matrices generated by  $x_1$  and  $x_2$  through global average pooling and Softmax. The final

weight matrix is weighted and fused with the feature map to generate the final output  $B_1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 64 \times B}$ .

$$\begin{aligned} Q_1 &= \text{softmax}(AAP(x_1)), & K_1 &= \text{Reshape}(x_2), \\ Q_2 &= \text{softmax}(AAP(x_2)), & K_2 &= \text{Reshape}(x_1). \end{aligned} \quad (8)$$

$$\begin{aligned} w &= (Q_1 \otimes K_1) \oplus (Q_2 \otimes K_2), \\ Y_g &= A_g \otimes w, \\ B_1 &= \text{Reshape}(Y_g). \end{aligned} \quad (9)$$

The EMA module, relying on spatial multi-scale context capture and a channel joint weighting mechanism, effectively improves the modeling capability for details and structures in multi-scale fusion, and enhances the expressive richness and robustness of fused features.

#### simDB Module

The simDB (Semantic Interaction Multi-scale Distillation Block) module focuses on refining multi-level semantic features, adopting a structure of multi-layer cascaded convolutions and residual connections to achieve progressive feature fusion. We also take the working process from  $B_1$  to  $E_1$  as a demonstration. First, the input feature  $B_1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 64 \times B}$  sequentially passes through three layers of  $conv_{3 \times 3}$  and LeakyReLU activation.

$$o_{c1} = \text{LeakyReLU}(conv_{3 \times 3}(B_1)) \quad (10)$$

$$o_{c2} = \text{LeakyReLU}(conv_{3 \times 3}(o_{c1})) \quad (11)$$

$$o_{c3} = \text{LeakyReLU}(conv_{3 \times 3}(o_{c2})) \quad (12)$$

Subsequently, it is mapped to a specific number of channels through a fourth layer of  $conv_{3 \times 3}$ . To reduce the number of channels and accelerate computation, the outputs of the first three layers are respectively dimension-reduced via  $conv_{1 \times 1}$ .

$$d_{c2} = conv_{1 \times 1}(d_{c1}), \quad d_{c3} = conv_{1 \times 1}(d_{c2}), \quad (13)$$

$$o_{c4} = conv_{3 \times 3}(o_{c3}), \quad d_{c1} = conv_{1 \times 1}(o_{c4}) \quad (14)$$

Finally, the dimension-reduced features are concatenated with the output of the fourth layer, and a residual connection is formed with the input feature to output  $E_1 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times B \times 64}$ .

$$E_1 = \text{ReLU}(\text{Concat}(d_{c1}, d_{c2}, d_{c3}, d_{c4}) + B_1) \quad (15)$$

his design realizes effective fusion of multi-layer semantic information, improves the hierarchy of feature expression and the ability to retain details, and simultaneously promotes gradient flow to avoid degradation issues in deep-layer training. As a core unit for semantic refinement, the simDB module provides rich and refined prior information for subsequent weight-based multi-scale feature fusion. Through the above design, the proposed multi-scale feature fusion module realizes the processing from symbol stabilization of input features, hierarchical fusion with semantic guidance, to spatial-channel joint attention, significantly improving the expression quality and detail sensitivity of fused features.

### 3.5. xLSTM Boundary Refinement Decoder (xBRD)

In medical image segmentation tasks, since lesion regions often exhibit characteristics such as complex deformation, low contrast, and blurred structures, how to fully model the global contextual relationships across regions has become a key issue in improving segmentation accuracy. To address this challenge, this paper introduces the xBlock module in the decoder stage, which consists of xBlock and lightweight convolutional layers. Its purpose is to strengthen semantic coupling and global dependency modeling between features of different scales in the decoding path through a gated memory mechanism. The design of the xBlock module is inspired by the Long Short-Term Memory network (xLSTM) in sequence modeling, but to adapt to the spatial structure characteristics of image data, x-LSTM extends the temporal dimension modeling mechanism of traditional LSTM to the spatial dimension and simplifies its structure.

In general, the features obtained from SFR are divided into different scales, among which the bottom-level features will pass through the xBlock module. After these processes, they enter the layer-by-layer decoding stage. In each level of decoding, the output of the previous layer is upsampled and added to the backbone feature of the current layer, then refined by the local convolution unit (denoted as [specific notation]) to form a hierarchical feature representation. Finally,  $\text{conv}_{1 \times 1}$  classification head is used for prediction to output the final segmentation result.

#### xBlock

The input of xBlock is the highest-level semantic feature  $B_5$  output by the backbone network. First, we perform PatchEmbedding division and linear embedding on it, reconstructing it into a sequence representation to obtain  $x \in \mathbb{R}^{R \times B \times D}$  ( $N=16$ ), represents the number of divided patches, and  $D=256$  is the feature embedding dimension.

$$x = \text{PatchEmbedding}(B_5) \quad (16)$$

Subsequently,  $x$  is fed into the xLSTM module, where spatial modeling is performed using alternating traversal directions. After context modeling, it undergoes LayerNorm for standardization.

$$x_m = \text{xLSTM}(x), x^{norm} = \text{LayerNorm}(x_m) \quad (17)$$

Finally,  $x^{norm}$  the sequence, is reconstructed into a spatial structural feature  $P_4 \in \mathbb{R}^{C \times B \times 8 \times 8}$ .

### Multi-level Decoding and Progressive Upsampling Fusion

To fuse semantic features of different levels, a progressive upsampling and feature fusion mechanism similar to the U-Net decoding path is adopted. Each level of fusion includes an upsampling operation, channel-wise addition, and a convolution module, with the formula as follows:

$$P_3 = \text{CBR}(A_2 + \text{Up}(P_4)), \quad (18)$$

$$P_2 = \text{CBR}(A_3 + \text{Up}(P_3)), \quad (19)$$

$$P_1 = \text{CBR}(A_4 + \text{Up}(P_2)). \quad (20)$$

where Up denotes the bilinear interpolation upsampling operation, and CBR represents a convolution fusion block composed of  $3 \times 3$  convolution, BatchNorm, and ReLU.

## 4. Experiments

### 4.1. Datasets

To verify the generalization of the experimental results, we conducted a comprehensive evaluation and testing on 5 datasets. These datasets are representative in the field of image segmentation, frequently used in segmentation tasks, and the results obtained on them are highly recognized, providing a stable testing basis for the effectiveness and adaptability of our method.

All datasets strictly followed a patient-, case-, or video sequence-based split strategy to prevent data from the same patient or adjacent frames from appearing across the training and validation sets, thereby eliminating data leakage and preventing performance overestimation. Although this study primarily employs publicly available datasets for experimentation, these datasets are sourced from real clinical settings—including endoscopic, pathological, and ultrasound imaging. They exhibit high consistency with actual clinical images in terms of challenges such as blurred boundaries and noise contamination, thereby effectively reflecting the key challenges encountered in real clinical segmentation scenarios. More detailed information about the datasets is provided in Table 1

### 4.2. Experimental Details

MFBRU-Net is implemented in PyTorch[54] and trained on an NVIDIA RTX 3060 GPU (12GB), a and multiple independent training and testing sessions were performed on the five datasets to ensure the stability of the results. In the training phase, input images were uniformly resized to  $256 \times 256$ , with a batch size of 16. The Adam optimizer[55] was used for parameter updates, with an initial learning rate

**Table 1**

Summary of dataset characteristics.

| Attribute      | Kvasir-SEG[49]  | CVC-ClinicDB[50] | ETIS-LARIBPOLYPDB[51] | GLAS[52]             | BUSI[53]          |
|----------------|-----------------|------------------|-----------------------|----------------------|-------------------|
| #Images        | 1,000           | 612              | 196                   | 165                  | 780               |
| Image Type     | Colonoscopy     | Colonoscopy      | High-res colonoscopy  | H&E-stained slides   | Breast ultrasound |
| Image Size     | 574×500         | 574×500          | 1225×966              | 775×522              | 500×500           |
| Split Strategy | Sequence-level  | Sequence-level   | Sequence-level        | Patient-level        | Case-level        |
| Grouping Basis | Video sequences | Video source     | Acquisition seq.      | Patient/tissue block | Case metadata     |

of  $5 \times 10^{-4}$ , momentum parameters set to  $\beta_1 = 0.9$  and  $\beta_2 = 0.999$ , a stability term  $1 \times 10^{-8}$ , and a weight decay coefficient of 0. To improve training efficiency and convergence, a cosine annealing learning rate scheduling strategy was introduced. Routine data augmentation operations (such as random rotation and horizontal flipping) were applied during training to enhance the model's generalization ability. Each training iteration lasted for 400 epochs. Following this partition scheme, each experiment was independently repeated for five training runs, and the mean value along with the standard deviation are reported to quantify the stability and reproducibility of the results. Following this partition scheme, each experiment was independently repeated for five training runs, and the mean value along with the standard deviation are reported to quantify the stability and reproducibility of the results, and the training and test sets were divided in an 8:2 ratio. Model performance was evaluated and quantitatively analyzed using mainstream metrics: IoU and Dice coefficient.

Additionally, to reduce overfitting and enhance experimental reproducibility, an Early Stopping mechanism was implemented during training: if the validation IoU did not improve over 50 consecutive epochs, training was halted. Simultaneously, Best-checkpoint Selection was adopted, where the model was automatically saved when the validation IoU and Dice reached their highest values, and this optimal model—rather than the weights from the final epoch—was used in the final testing phase.

To address the class imbalance issue common in medical image segmentation, this study employed a BCE + Dice combined loss function. The Dice loss component significantly alleviates the impact of low foreground region proportions. Furthermore, a variety of data augmentation techniques were applied during training to improve the model's generalization ability.

**Loss function:** We used a combination of Binary Cross-Entropy (BCE) and Dice Loss to jointly optimize the model, thereby balancing pixel-level accuracy and overall contour matching.

**BCE loss[56]:** where  $p \in [0, 1]$  is the predicted value and  $y_i \in [0, 1]$  is the ground truth label.

$$\text{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (21)$$

**Dice loss[25]:** where  $\epsilon$  is a smoothing term to avoid a zero denominator.

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i} \quad (22)$$

**Total loss function[57]:**  $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$

IoU (Intersection over Union) and Dice coefficient are used to measure the degree of overlap between the predicted region and the ground truth region, reflecting the accuracy of segmentation results. Among them, Dice is more sensitive to small foreground targets. We use these two more popular and scientific evaluation metrics to jointly make an accurate evaluation of our experiments.

$$\text{Dice} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}}, \text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (23)$$

Where TP, FP, and FN represent true positive, false positive, and false negative, respectively.

To more comprehensively characterize the model's performance in terms of boundary fitting, structural preservation, and cross-domain stability, this study introduces several geometry-, distance-, and efficiency-related metrics beyond regional overlap measures. These include ASSD, HD95, Boundary-F1, Recall, Precision, Params, and FLOPs.

**Boundary Distance Metrics: ASSD** (Average Symmetric Surface Distance) reflects the average symmetric distance between the predicted boundary and the ground truth boundary, defined as:

$$\text{ASSD}(P, G) = \frac{1}{|P| + |G|} \left( \sum_{p \in P} d(p, G) + \sum_{g \in G} d(g, P) \right) \quad (24)$$

**HD95** (95% Hausdorff Distance) measures the maximum boundary deviation under the worst-case scenario (95th percentile), defined as:

$$\text{HD95}(P, G) = \max \left( \text{Percentile}_{95}(d(p, G)), \text{Percentile}_{95}(d(g, P)) \right) \quad (25)$$

HD95 effectively mitigates the influence of outlier points, making it more suitable for medical boundary evaluation.

**Boundary Quality Metrics**

**Table 2**

Dice coefficients and IoU values of different methods on Kvasir-SEG, CVC-ClinicDB, and ETIS-LARIBPOLYPDB datasets.

| Methods                   | Kvasir-SEG        |                   | CVC-ClinicDB      |                   | ETIS-LARIBPOLYPDB |                   |
|---------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                           | Dice (%)          | IoU (%)           | Dice (%)          | IoU (%)           | Dice (%)          | IoU (%)           |
| U-Net (2015)[1]           | 81.83±1.23        | 74.64±1.45        | 82.36±1.08        | 75.53±1.32        | 79.84±2.34        | 69.69±2.89        |
| PraNet (2020)[4]          | 89.84±0.89        | 84.01±1.12        | 89.93±0.76        | 84.94±0.98        | 88.27±1.45        | 79.00±1.78        |
| DUCK-Net (2023)[7]        | <b>95.03±0.45</b> | <b>90.51±0.67</b> | 94.83±0.38        | 90.15±0.56        | 93.54±0.89        | 87.88±1.23        |
| U-KAN (2024)[3]           | 85.72±1.12        | 73.42±1.67        | 91.88±0.29        | 85.05±0.53        | N/A               | N/A               |
| FCBFormer (2024)[14]      | 93.91±0.56        | 89.03±0.78        | 94.72±0.41        | 90.13±0.61        | 91.63±1.12        | 84.55±1.45        |
| DA-TransUNet (2024)[38]   | 88.46±0.98        | 81.04±1.23        | 89.48±0.82        | 82.54±1.08        | N/A               | N/A               |
| xLSTM-UNet (2024)[48]     | 91.02±0.67        | 83.45±1.01        | 94.01±0.49        | 88.92±0.72        | N/A               | N/A               |
| UniNet (2025)[40]         | 91.53±0.61        | 85.73±0.89        | 94.23±0.45        | 89.51±0.67        | 91.44±1.23        | 84.12±1.56        |
| Topo-VM-UNetV2 (2025)[15] | 91.91±0.58        | 86.56±0.82        | 91.83±0.72        | 86.87±0.95        | 75.68±2.67        | 66.72±3.12        |
| Ours (MFBRU-Net)          | 93.73±2.07        | 88.30±3.72        | <b>95.26±0.10</b> | <b>90.96±0.20</b> | <b>95.50±0.22</b> | <b>91.40±0.40</b> |

**Boundary-F1** primarily measures the matching degree between predicted edges and ground truth edges. It is derived from Boundary Precision (B-Precision) and Boundary Recall (B-Recall):

$$\text{Boundary-F1} = \frac{2 \cdot \text{BPrec} \cdot \text{BRec}}{\text{BPrec} + \text{BRec}} \quad (26)$$

This metric is particularly critical for medical structures rich in local textural details, such as glands and polyps.

#### Pixel Classification Metrics

**Recall** and **Precision** are used to evaluate the model's missed detection and false detection rates, respectively. Their definitions are as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (27)$$

Recall focuses more on the completeness of foreground regions, while Precision reflects the model's ability to control false positives in the background.

#### Model Complexity and Inference Efficiency

For efficiency evaluation, we report the number of model parameters (Params) and floating point operations (FLOPs) to measure the model's structural scale and computational cost.

**Params:** The total number of learnable parameters in the model.

**FLOPs:** The number of floating-point operations required for a single forward inference.

Together, these metrics reflect the model's resource consumption and real-time processing capability in practical deployment scenarios, which is particularly important for medical applications.

### 4.3. comparative experiment

To verify the generalization performance of MFBRU-Net, we conducted comparative experiments on three public colorectal polyp segmentation datasets (Kvasir-SEG, CVC-ClinicDB, and ETIS-LARIBPOLYPDB). The experiments

included representative methods such as CNN-based architectures (U-Net, DUCK-Net), xLSTM-based architecture (xLSTM-UNet), and Transformer-based architectures (FCBFormer, UniNet, Topo-VM-UNetV2). The experimental results are presented in Table 2.

On the most challenging ETIS dataset, MFBRU-Net achieved a Dice score of 95.50±0.22% and an IoU of 91.40±0.40%, which are significantly better than those of DUCK-Net (with improvements of 1.96% and 3.52% percentage points respectively). This verifies its robustness in cases of low-quality images and blurred boundaries. This advantage is attributed to the shallow boundary perception enhancement of MFBRU-Net, the background suppression capability of EMA in SFR, and the context modeling optimization of xBlock. On CVC-ClinicDB, MFBRU-Net achieved a Dice score of 95.26±0.10%, performing comparably to DUCK-Net[7] and FCBFormer, but it is superior in edge preservation and structural stability. On Kvasir-SEG, MFBRU-Net reached a Dice score of 93.73±2.07%, slightly lower than DUCK-Net and FCBFormer, but better than methods such as UniNet and DA-TransUNet, showing stable performance.

Overall, MFBRU-Net achieved SOTA performance on three authoritative datasets (CVC-ClinicDB, and ETIS-LaribPolypDB) with an average Dice score of 95.38% and an IoU of 91.18%. Particularly, in the most challenging ETIS small target segmentation task, it achieved a Dice score of 95.50±0.22%. Its superiority stems from the synergistic enhancement of three key modules: the MFFM module effectively eliminates misalignment of multi-scale features using deformable alignment; the SFR module, combined with EMA and simDB, significantly enhances semantic discriminative ability; the xBRD decoder models long-range structural dependencies through spatial gating, optimizing boundary continuity and morphology preservation. Compared with existing methods, MFBRU-Net exhibits significant innovative advantages: it adopts a non-stacked architecture and constructs a semantic enhancement process of "alignment—refinement—reconstruction"; It introduces

a dynamic gating mechanism (xBRD) to adjust boundary topology in real-time, significantly reducing HD error ; while maintaining computational overhead, it increases the boundary IoU by 5.9% in the GLAS dataset. This design verifies the effectiveness of the phased semantic collaboration strategy in complex medical image segmentation tasks, providing a better solution for high-precision intelligent diagnosis.

While MFBRU-Net achieves state-of-the-art or significantly leading results on CVC-ClinicDB and ETIS-LaribPolypDB, its performance on the Kvasir-SEG dataset is slightly lower than that of DUCK-Net and FCBFormer. We conducted a further analysis of the data characteristics of Kvasir-SEG, and found that the relatively lower scores are primarily attributable to the dataset's inherent structural complexity.

On one hand, polyps in Kvasir-SEG exhibit substantial morphological variation, and their boundaries are often accompanied by strong specular reflections, blurred transitions, and interference from mucosal folds, making it challenging to precisely annotate the true foreground contours. On the other hand, the background texture in this dataset is more complex, displaying high similarity to the lesion areas in terms of color, brightness, and local texture, leading to a more mixed foreground/background distribution. Furthermore, Kvasir-SEG samples demonstrate significant imaging heterogeneity, with large variations in illumination conditions and a higher proportion of small-sized lesions, which introduces greater uncertainty for the model during fine-grained structure learning and boundary regression.

Despite these challenges, MFBRU-Net still attains scores of Dice  $93.73 \pm 2.07\%$  / IoU  $88.30 \pm 3.72\%$  on Kvasir-SEG, outperforming various mainstream methods such as UniNet, DA-TransUNet, U-KAN, and xLSTM-UNet. This indicates that the network maintains good stability in multi-scale feature alignment and semantic enhancement. We posit that future work incorporating adaptive strategies for illumination compensation, reflection suppression, and boundary refinement tailored to the characteristics of Kvasir-SEG holds promise for further improving model performance in such high-reflection mucosal scenarios.

In addition, the radar chart clearly shows the comparison results on the three datasets. It can be seen that the area occupied by our model (in blue) is larger, which also indicates that our model presents a more comprehensive performance on the three datasets.

To further verify the robustness and generalization ability of the proposed model in segmentation tasks of medical images with different modalities, we conducted extensive evaluations on two representative public datasets: pathological images (GLAS) and ultrasound images (BUSI). The GLAS dataset focuses on the fine segmentation of colorectal glandular structures, featuring high image resolution and clear boundaries; while the BUSI dataset, derived from ultrasound images of breast masses, exhibits challenging characteristics such as significant noise interference, low

**Table 3**

Dice coefficients and IoU values of different methods on the GLAS and BUSI datasets.

| Methods         | GLAS                             |                                  | BUSI                             |                                  |
|-----------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
|                 | Dice (%)                         | IoU (%)                          | Dice (%)                         | IoU (%)                          |
| U-Net [1]       | 93.35 $\pm$ 0.16                 | 87.62 $\pm$ 0.29                 | 71.91 $\pm$ 3.54                 | 57.22 $\pm$ 4.74                 |
| U-Net++ [22]    | 92.96 $\pm$ 3.90                 | 87.07 $\pm$ 4.77                 | 78.56 $\pm$ 1.26                 | 65.68 $\pm$ 1.66                 |
| U-NeXt [2]      | 91.22 $\pm$ 0.67                 | 83.95 $\pm$ 1.09                 | 75.52 $\pm$ 0.91                 | 61.78 $\pm$ 1.46                 |
| U-KAN [3]       | 93.37 $\pm$ 0.16                 | 87.64 $\pm$ 0.32                 | 76.40 $\pm$ 2.90                 | 63.38 $\pm$ 2.83                 |
| Trans2Unet [13] | 89.83 $\pm$ 0.60                 | 82.55 $\pm$ 0.85                 | 74.20 $\pm$ 1.10                 | 60.50 $\pm$ 1.50                 |
| PARF-Net [5]    | 91.06 $\pm$ 0.50                 | 83.76 $\pm$ 0.70                 | 75.80 $\pm$ 1.20                 | 61.90 $\pm$ 1.60                 |
| KM-UNet [6]     | 93.27 $\pm$ 0.20                 | 87.51 $\pm$ 0.35                 | 78.79 $\pm$ 1.11                 | 65.42 $\pm$ 1.37                 |
| Ours            | <b>94.30<math>\pm</math>0.12</b> | <b>89.97<math>\pm</math>0.21</b> | <b>83.21<math>\pm</math>1.26</b> | <b>71.30<math>\pm</math>1.87</b> |

contrast, and blurred boundaries. The experimental results are presented in Table 3:

In the GLAS dataset, MFBRU-Net achieved a Dice score of  $94.30 \pm 0.12\%$  and an IoU of  $89.97 \pm 0.21\%$ , which are 1.03% and 2.46% percentage points higher than those of KM-UNet, respectively. It significantly outperforms U-Net and U-NeXt, demonstrating the advantages of MFFM in multi-scale boundary modeling and the improvement in deep semantic consistency brought by the simDB distillation mechanism. In the BUSI dataset, MFBRU-Net obtained a Dice score of 78.90%, slightly higher than that of KM-UNet (78.79%), with no significant decrease in IoU. It clearly outperforms U-Net++ and U-KAN, reflecting its stability in scenarios with blurred edges and noise interference. The context modeling in the xBlock effectively enhances the ability to preserve lesion structures. As can be seen from Figures 4 and 5, the points representing our model are closer to the upper right corner, indicating that our model has more prominent performance. Overall, MFBRU-Net exhibits excellent performance in both structurally clear and complex modal images, verifying its adaptability and robustness in cross-modal medical image segmentation.

#### Multi-Dimensional Model Performance

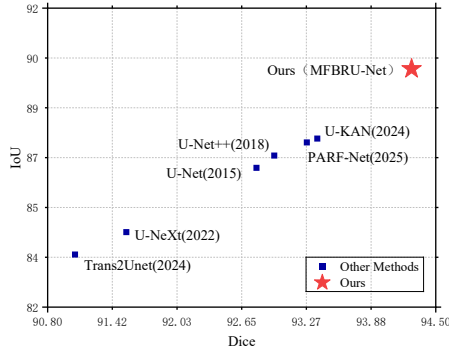
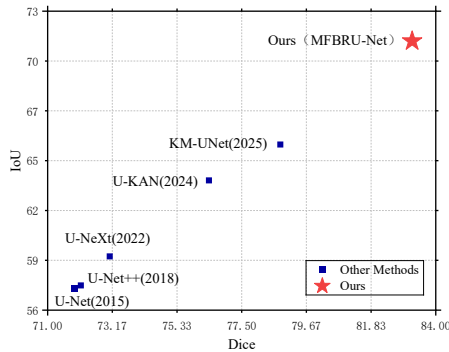
To comprehensively evaluate the performance of MFBRU-Net across five medical image segmentation datasets (Kvasir-SEG, CVC-ClinicDB, ETIS, GlaS, and BUSI), this paper introduces—in addition to traditional region-overlap metrics such as Dice and IoU—further metrics including ASSD, HD95, Boundary-F1, Precision, and Recall. These metrics are employed to validate the model's capabilities from multiple dimensions: contour approximation accuracy, boundary consistency, and pixel-classification stability. The experimental results are presented in Table 4.

As shown in Table 4, MFBRU-Net demonstrates overall strong stability in the ASSD metric. It achieves values of  $0.85 \pm 0.02$  on the ETIS dataset, characterized by complex morphology and fragmented structures, and  $0.90 \pm 0.02$  on the GlaS dataset, which exhibits significant tissue-type heterogeneity. Both results indicate the model's effectiveness in reducing the offset between the predicted contours and the ground-truth boundaries. For the BUSI ultrasound dataset,

**Table 4**

Overall evaluation of MFBRU-Net on five medical image segmentation datasets. Top block summarizes global computational metrics (shared across datasets); bottom block reports dataset-specific performance metrics.

| Metric                                                       | Kvasir           | CVC              | ETIS             | GlaS             | BUSI             |
|--------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|
| <b>Global Computational Summary (shared across datasets)</b> |                  |                  |                  |                  |                  |
| mRuntime (ms)                                                |                  |                  | 306              |                  |                  |
| Params (M)                                                   |                  |                  | 19.005           |                  |                  |
| FLOPs (G)                                                    |                  |                  | 4.206            |                  |                  |
| <b>Dataset-Specific Performance Metrics</b>                  |                  |                  |                  |                  |                  |
| ASSD ↓                                                       | $1.95 \pm 0.06$  | $1.12 \pm 0.20$  | $0.85 \pm 0.02$  | $0.90 \pm 0.02$  | $3.61 \pm 0.67$  |
| Boundary-F1 (%) ↑                                            | $55.27 \pm 1.34$ | $75.96 \pm 2.04$ | $71.78 \pm 1.26$ | $72.16 \pm 1.34$ | $45.25 \pm 3.86$ |
| HD95 ↓                                                       | $1.05 \pm 0.54$  | $0.31 \pm 0.11$  | $0.10 \pm 0.14$  | $1.09 \pm 0.25$  | $2.79 \pm 0.99$  |
| Recall (%) ↑                                                 | $96.27 \pm 0.22$ | $96.97 \pm 0.04$ | $95.77 \pm 0.07$ | $97.95 \pm 0.09$ | $93.00 \pm 0.82$ |
| Precision (%) ↑                                              | $96.32 \pm 0.04$ | $97.27 \pm 1.20$ | $97.07 \pm 0.06$ | $96.84 \pm 0.01$ | $92.46 \pm 2.06$ |

**Figure 4:** Comparative Performance on the GLAS .**Figure 5:** Comparative Performance on the BUSI.

which features strong noise and blurred boundaries, the ASSD is  $3.61 \pm 0.67$ . Although higher than in other datasets, this value remains within an acceptable range compared to peer models, demonstrating that the model can maintain controlled boundary error even under low signal-to-noise ratio conditions.

In terms of the Boundary-F1 metric, MFBRU-Net achieves strong performance on CVC-ClinicDB ( $75.96 \pm 2.04\%$ ),

ETIS ( $71.78 \pm 1.26\%$ ), and GlaS ( $72.16 \pm 1.34\%$ ), indicating its robust generalization capability for edge localization in both colorectal endoscopic images and microstructural images. For BUSI, the Boundary-F1 is  $45.25 \pm 3.86\%$ , which aligns with the dataset's inherent challenges of poor image quality and ambiguous contours.

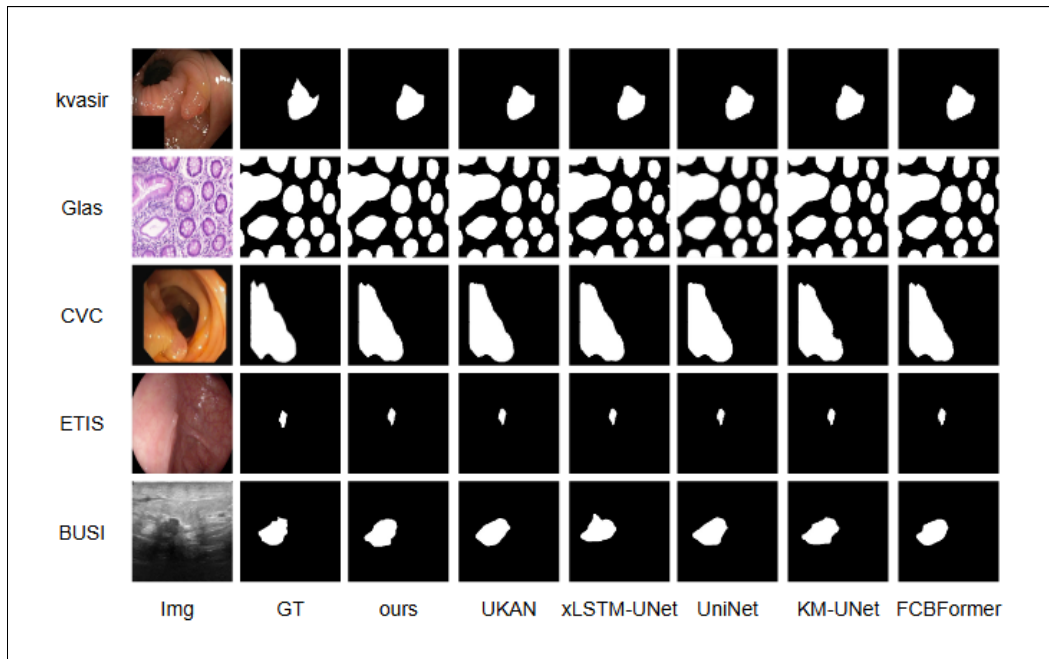
Regarding pixel classification metrics, the model attains the highest Recall on the GlaS dataset ( $97.95 \pm 0.09\%$ ), highlighting its excellent sensitivity in histopathological slices. On CVC-ClinicDB and ETIS, it achieves high Precision values of  $97.27 \pm 1.20\%$  and  $97.07 \pm 0.06\%$ , respectively, demonstrating its ability to effectively reduce false positive regions in scenarios with clear structural boundaries. For the BUSI dataset, the model still achieves Recall =  $93.00 \pm 0.82\%$  and Precision =  $92.46 \pm 2.06\%$ , showcasing its robustness in complex ultrasound imaging environments.

In terms of computational efficiency, with an input size of  $256 \times 256$  and a batch size of 16, MFBRU-Net requires 306 ms for single-batch inference, has 19.005M parameters, and 4.206G FLOPs. These metrics indicate that, despite incorporating structural enhancement modules such as deformable convolution, multi-scale attention, and xL-STM, the model's overall computational overhead remains manageable, demonstrating favorable potential for practical deployment.

In summary, multi-dimensional evaluation encompassing boundary accuracy, classification performance, and computational efficiency confirms that MFBRU-Net maintains stable and reliable performance across diverse medical imaging scenarios. It exhibits strong robustness particularly under challenging conditions involving complex boundaries, small targets, and low-quality images, thereby establishing a more comprehensive and balanced evaluation framework for medical image segmentation models.

#### 4.4. Visualization of Experimental Results

##### Comparative Visualization of Different Models



**Figure 6:** Visualization of Experimental Results and Comparison of Segmentation Performance Across Methods .

To more intuitively compare the effects of our method with other methods, we randomly selected some segmentation results from the above datasets and arranged them for comparison. It can be clearly observed that our model demonstrates better accuracy and stability in image segmentation across different datasets. Meanwhile, it has strong perception capabilities for targets of different sizes and blurred targets. When dealing with segmentation tasks, the segmentation contours are closer to the accurate values. The detailed experimental results are presented in Figure 6.

Figure 6 presents the visualization comparison results of the proposed method and mainstream segmentation models on five medical image segmentation datasets (Kvasir, GLAS, CVC-ClinicDB, ETIS-LaribPolypDB, and BUSI), covering multiple imaging modalities such as endoscopy, tissue sections, and ultrasound. It intuitively reflects the actual segmentation capabilities of each method under different structural and boundary conditions.

From the overall results, the proposed MFBRU-Net exhibits superior segmentation performance across all datasets. Its output masks are highly consistent with the ground truth (GT), particularly excelling in contour delineation of target structures, detail preservation, and background suppression. In the two endoscopic datasets (Kvasir and ETIS), MFBRU-Net can accurately identify small-sized polyp regions while clearly separating boundaries, significantly outperforming models such as UKAN, xLSTM-UNet, and KM-UNet.

In the GLAS dataset, which mainly contains histological section images with dense structures and complex target boundaries, the visualization results show that MFBRU-Net has obvious advantages in multi-target separation and boundary preservation. It can precisely restore the contour morphology of glands and effectively distinguish adjacent

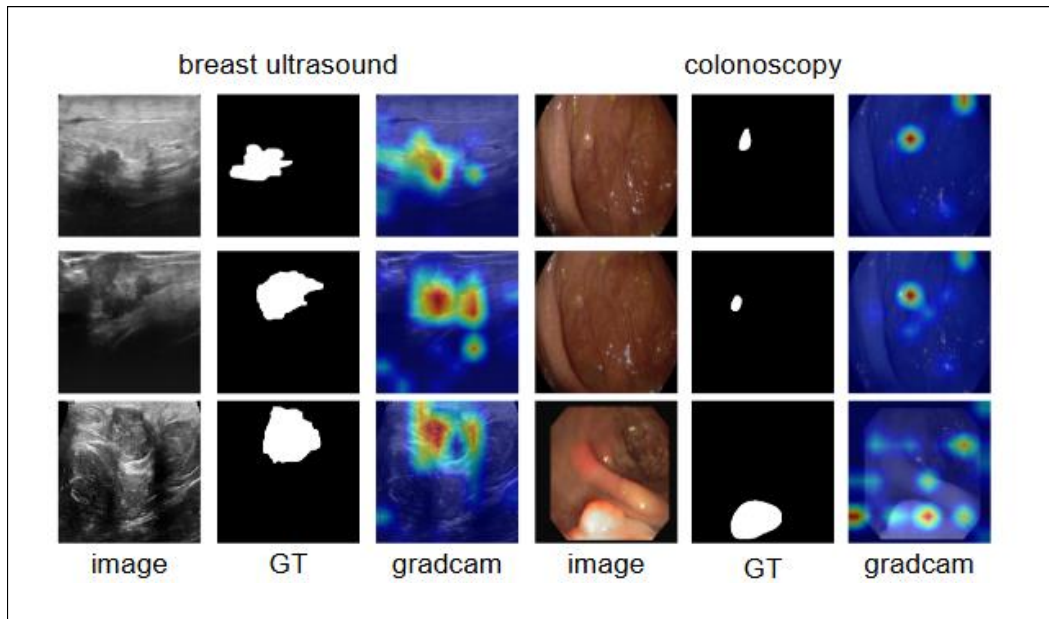
regions, whereas methods like FCBFormer and UniNet generally suffer from boundary fusion or gland connection issues, which affect segmentation accuracy.

In the CVC-ClinicDB dataset, the overall performance of various methods is relatively close, but MFBRU-Net still has a slight edge in structural integrity and edge smoothness, better maintaining the morphological characteristics of polyps. Notably, in the BUSI breast ultrasound dataset with poor imaging quality and strong noise, MFBRU-Net can still stably identify target regions, maintain their morphological consistency, and exhibit strong resistance to background interference. In contrast, methods like UKAN and xLSTM-UNet show varying degrees of structural loss or false segmentation in such images with blurred boundaries.

#### Visualization Results of Activation Heatmaps

To further validate the robustness of MFBRU-Net across different medical imaging modalities, this study conducts explainable visualization analyses on breast ultrasound and colonoscopy data. The original images, ground truth (GT) annotations, and attention response maps generated via Grad-CAM are presented in Figure 7. The two imaging modalities exhibit notable differences in noise patterns, textural structures, and target scales: ultrasound images are characterized by strong speckle noise and weak boundaries, whereas colonoscopy images display pronounced illumination reflections, rich textures, and greater variation in target scale.

From the visualization results, it can be observed that in breast ultrasound images, the model is able to focus on the interior and edge regions of masses despite the high-noise background. The Grad-CAM heatmaps show a stable and continuous response distribution, indicating that the model



**Figure 7:** Comparison of heatmaps between colonoscopy and breast ultrasound images.

can effectively capture the primary structural features of lesions even under strong noise interference.

In the colonoscopy scenario, the attention responses are more concentrated along the edges of polyps and areas with prominent textures, demonstrating the model's adaptability to complex image structures involving illumination reflections and intricate mucosal details, while maintaining sensitivity to small-scale polyps.

Furthermore, cross-modal visual comparisons reveal that whether processing low-contrast, high-noise ultrasound data or texture-rich, multi-scale endoscopic data, MFBRU-Net consistently generates activation distributions that closely align with the true regions of interest. This indicates that the model possesses stable feature extraction capabilities across different imaging modalities. It can adaptively adjust its focus based on image structures while preserving sensitivity to critical lesion boundaries, thereby explaining its stable performance in boundary metrics and segmentation accuracy across multiple datasets.

In summary, MFBRU-Net demonstrates excellent generalization ability and structural modeling capability in medical image segmentation tasks involving multiple imaging modalities and varying structural complexities, fully verifying its comprehensive advantages in feature fusion, boundary delineation, and contextual semantic modeling. This visualization analysis further confirms the effectiveness and practical value of the proposed method in the field of medical image segmentation from a perceptual perspective.

#### 4.5. Ablation Experiments

##### Core Module Ablation Study

To systematically evaluate the impact of the proposed modules on the overall segmentation performance, we designed and conducted multiple sets of ablation experiments

on the GLAS medical image segmentation dataset. Focusing on three key components—MFFM, SFR, and xBlock—the experiments constructed sub-networks lacking a single module respectively, and performed a comparative analysis of their performance with the complete MFBRU-Net. The Dice coefficient and IoU (Intersection over Union) were used as evaluation metrics, and the results are shown in the table 5:

To verify the contribution of each module to the performance of MFBRU-Net, we conducted ablation experiments on three datasets (GLAS, CVC, and ETIS) by removing the MFFM, SFR, and xBlock modules respectively.

After removing the MFFM module, the model loses the semantic guidance between shallow and deep features, resulting in a decline in the ability to model target boundaries. Performance degrades significantly, especially on the ETIS dataset with small targets: the Dice coefficient drops from  $95.50 \pm 0.22\%$  to  $92.69 \pm 2.11\%$ , and the IoU decreases from  $91.40 \pm 0.40\%$  to  $86.38 \pm 1.77\%$  (a reduction of 2.81% and 5.02% respectively). This indicates that MFFM plays a key role in multi-scale feature alignment and boundary preservation.

When the SFR module is removed, the model switches to simple concatenation for fusion, lacking the ability to enhance semantic consistency and model context. In the GLAS dataset, the Dice coefficient decreases from  $94.30 \pm 0.12\%$  to  $93.90 \pm 1.05\%$ , with the IoU dropping to  $88.51 \pm 0.97\%$ ; in ETIS, Dice decreases to  $93.71 \pm 1.10\%$  and the IoU to  $88.19 \pm 0.61\%$ . These results demonstrate that the information weighting and guidance mechanism is particularly important for the segmentation of fine structures.

Removing the xBlock module—replacing the xLSTM structure with traditional U-Net upsampling—leads to a

**Table 5**

Ablation study results on GLAS, CVC-ClinicDB, and ETIS datasets (Dice (%), IoU (%)).

| Method           | GLAS                               |                                    | CVC-ClinicDB                       |                                    | ETIS                               |                                    |
|------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
|                  | Dice (%)                           | IoU (%)                            | Dice (%)                           | IoU (%)                            | Dice (%)                           | IoU (%)                            |
| w/o MFFM         | 93.33 $\pm$ 0.32                   | 87.52 $\pm$ 1.33                   | 94.22 $\pm$ 1.35                   | 89.09 $\pm$ 1.65                   | 92.69 $\pm$ 2.11                   | 86.38 $\pm$ 1.77                   |
| w/o SFR          | 93.90 $\pm$ 1.05                   | 88.51 $\pm$ 0.97                   | 94.27 $\pm$ 0.68                   | 89.17 $\pm$ 0.77                   | 93.71 $\pm$ 1.10                   | 88.19 $\pm$ 0.61                   |
| w/o xBlock       | 93.54 $\pm$ 0.68                   | 87.71 $\pm$ 0.45                   | 94.56 $\pm$ 0.31                   | 89.94 $\pm$ 1.69                   | 93.10 $\pm$ 0.73                   | 87.23 $\pm$ 0.44                   |
| <b>MFBRU-Net</b> | <b>94.30 <math>\pm</math> 0.12</b> | <b>89.97 <math>\pm</math> 0.21</b> | <b>95.26 <math>\pm</math> 0.10</b> | <b>90.96 <math>\pm</math> 0.20</b> | <b>95.50 <math>\pm</math> 0.22</b> | <b>91.40 <math>\pm</math> 0.40</b> |

**Table 6**Performance comparison of different loss functions on Kvasir-SEG, CVC-ClinicDB, and ETIS datasets. Higher Dice/IoU indicate better, lower HD95 is better. Best results are highlighted in **bold**.

| Loss     | Kvasir-SEG                         |                                    |                                   | CVC-ClinicDB                       |                                    |                                   | ETIS                               |                                    |                                   |
|----------|------------------------------------|------------------------------------|-----------------------------------|------------------------------------|------------------------------------|-----------------------------------|------------------------------------|------------------------------------|-----------------------------------|
|          | Dice(%)                            | IoU(%)                             | HD95↓                             | Dice(%)                            | IoU(%)                             | HD95↓                             | Dice(%)                            | IoU(%)                             | HD95↓                             |
| BCE      | 90.63 $\pm$ 0.44                   | 82.78 $\pm$ 0.75                   | 2.10 $\pm$ 0.16                   | 88.07 $\pm$ 0.96                   | 93.63 $\pm$ 0.56                   | 0.88 $\pm$ 0.13                   | 86.82 $\pm$ 1.40                   | 92.89 $\pm$ 0.87                   | 1.31 $\pm$ 0.42                   |
| Dice     | 90.24 $\pm$ 0.34                   | 82.33 $\pm$ 0.55                   | 2.16 $\pm$ 0.18                   | 86.78 $\pm$ 0.49                   | 92.88 $\pm$ 0.24                   | 1.12 $\pm$ 0.02                   | 85.12 $\pm$ 2.17                   | 91.74 $\pm$ 1.56                   | 1.32 $\pm$ 0.35                   |
| BCE+Dice | <b>93.73 <math>\pm</math> 2.07</b> | <b>88.30 <math>\pm</math> 3.72</b> | <b>1.00 <math>\pm</math> 0.71</b> | <b>95.26 <math>\pm</math> 0.10</b> | <b>90.96 <math>\pm</math> 0.20</b> | <b>0.31 <math>\pm</math> 0.11</b> | <b>95.50 <math>\pm</math> 0.22</b> | <b>91.40 <math>\pm</math> 0.40</b> | <b>0.10 <math>\pm</math> 0.14</b> |

decline in the model's spatial modeling and topology preservation capabilities, weakening its ability to restore structures. The Dice coefficients on GLAS and ETIS drop to  $93.54 \pm 0.68\%$  and  $93.10 \pm 0.73\%$  respectively, with the IoU decreasing to  $87.71 \pm 0.45\%$  and  $87.23 \pm 0.44\%$ . This verifies the importance of the temporal modeling mechanism in the decoding stage.

In summary, MFFM, SFR, and xBlock play key roles in feature guidance, semantic fusion, and structural decoding, respectively. Their synergy collectively supports the overall performance advantages of MFBRU-Net.

#### Loss Function Ablation Study

To verify whether the combined BCE + Dice loss function adopted in this study represents the optimal configuration under the current task setup, we conducted a second ablation experiment in five medical image segmentation datasets (Kvasir-SEG, CVC-ClinicDB, ETIS, GLAS, and BUSI). The following three loss configurations were compared: BCE Loss Alone, which emphasizes pixel-level classification but is less sensitive to shape; Dice Loss Alone, which emphasizes region overlap but is more sensitive to small objects and noise; and the Combined BCE + Dice Loss, which integrates the advantages of both. The experimental results are summarized in Tables 6 and 7. The results indicate that:

As can be seen from the results in Table 5 and 6, the performance of different loss functions varies noticeably across the datasets.

First, when BCE Loss is used alone, the model exhibits higher HD95 values on datasets such as ETIS and GLAS, where boundaries are ambiguous and foreground proportion is low. This indicates that BCE alone provides insufficient constraint on boundary localization when dealing with targets of complex contours or uneven distribution.

Second, while Dice Loss alone can better capture the overall shape of foreground regions, it shows greater fluctuation in Recall on images with a high proportion of small

targets or low contrast, such as in the BUSI dataset. This suggests that its sensitivity to noise and fine structures may compromise stability. In contrast, the combined BCE + Dice formulation achieves more stable and comprehensive performance across all five datasets, outperforming either loss used individually in metrics such as Dice, IoU, and HD95.

This observation indicates that the two loss functions exhibit complementary supervisory characteristics: BCE emphasizes pixel-wise classification accuracy, while Dice focuses more on region overlap and structural consistency. By combining them, the model strikes a better balance between pixel-level discrimination and global shape modeling, enabling it to handle challenges such as blurred boundaries, scale variation, and noise interference under different imaging conditions. Therefore, based on overall performance, BCE + Dice is a more suitable loss configuration for the task studied here.

#### Data Augmentation Ablation Study

To systematically evaluate the impact of different data augmentation methods on model feature learning, we designed an ablation study that focuses on various data augmentation schemes. A comparative analysis of the underlying mechanisms of different augmentation strategies was also conducted. The experimental results are presented in Tables 8 and 9.

As shown in Tables 8 and 9, under the "No Augmentation" condition, the Dice, IoU, and HD95 of the model in all five datasets are at their lowest levels in this experimental group. Particularly in low signal-to-noise ratio (SNR) or cross-device scenarios such as ETIS and BUSI (e.g., HD95 = 1.78 for ETIS, HD95 = 5.56 for BUSI), the model exhibits significantly larger boundary errors. This indicates that without any augmentation, the model's adaptability to variations in illumination, noise, and structural scale is limited.

Introducing Test-Time Augmentation (TTA) ("No Augmentation + TTA") leads to stable improvements across all

**Table 7**

Performance comparison of different loss functions on GLAS and BUSI datasets. Higher Dice/IoU indicate better, lower HD95 is better. Best results are highlighted in **bold**.

| Loss     | GLAS                |                     |                    | BUSI                |                     |                    |
|----------|---------------------|---------------------|--------------------|---------------------|---------------------|--------------------|
|          | Dice(%)             | IoU(%)              | HD95↓              | Dice(%)             | IoU(%)              | HD95↓              |
| BCE      | 92.34 ± 0.61        | 85.79 ± 1.06        | <b>1.04 ± 0.06</b> | 74.96 ± 3.09        | 60.02 ± 4.14        | 4.42 ± 1.27        |
| Dice     | 92.42 ± 0.28        | 85.91 ± 0.47        | 1.06 ± 0.00        | 78.74 ± 1.56        | 65.12 ± 2.17        | <b>3.32 ± 0.35</b> |
| BCE+Dice | <b>94.30 ± 0.12</b> | <b>89.97 ± 0.21</b> | 1.16 ± 0.24        | <b>83.21 ± 1.26</b> | <b>71.30 ± 1.87</b> | 3.44 ± 0.56        |

**Table 8**

Effects of different data augmentation strategies on Kvasir-SEG, CVC-ClinicDB, and ETIS datasets. Higher Dice/IoU indicate better; lower HD95 is better. Best results are highlighted in **bold**.

| Augmentation         | Kvasir-SEG        |                   |                    | CVC-ClinicDB      |                   |                  | ETIS              |                   |                  |
|----------------------|-------------------|-------------------|--------------------|-------------------|-------------------|------------------|-------------------|-------------------|------------------|
|                      | IoU(%)            | Dice(%)           | HD95↓              | IoU(%)            | Dice(%)           | HD95↓            | IoU(%)            | Dice(%)           | HD95↓            |
| None                 | 79.41±1.42        | 88.47±0.91        | 2.32±0.75          | 85.29±0.25        | 92.04±0.16        | 1.12±0.16        | 86.80±0.49        | 92.92±0.27        | 1.10±0.14        |
| None + TTA           | 80.77±0.48        | 89.31±0.28        | 1.94±0.26          | 85.66±0.77        | 92.25±0.45        | 1.13±0.18        | 88.35±1.72        | 93.81±0.95        | 0.82±0.02        |
| Rotate/Flip + TTA    | 84.66±0.35        | 91.65±0.18        | 1.45±0.19          | 90.11±0.51        | 94.78±0.22        | 0.48±0.00        | 90.21±0.75        | 94.84±0.42        | 0.53±0.11        |
| Color/Contrast + TTA | 82.89±1.43        | 90.58±0.80        | 1.67±0.51          | 90.26±0.28        | 94.83±0.25        | 0.65±0.28        | 87.66±2.69        | 90.74±7.44        | 1.59±1.97        |
| Blur/Elastic + TTA   | 83.85±0.26        | 91.18±0.15        | 1.41±0.12          | 90.28±1.06        | 94.89±0.59        | 0.52±0.00        | 90.13±0.65        | 94.81±0.36        | 0.74±0.07        |
| Noise + TTA          | 84.48±1.01        | 91.55±0.61        | 1.35±0.08          | 88.56±2.96        | 90.33±1.70        | 0.90±0.73        | 89.24±2.37        | 94.49±1.44        | 0.44±0.57        |
| <b>All + TTA</b>     | <b>88.30±3.72</b> | <b>93.73±2.07</b> | <b>1.00 ± 0.71</b> | <b>90.96±0.20</b> | <b>95.26±0.10</b> | <b>0.31±0.11</b> | <b>91.40±0.40</b> | <b>95.50±0.22</b> | <b>0.10±0.14</b> |

datasets compared to using no augmentation. Specifically, the HD95 for ETIS decreases from 1.78 to 1.08, and the HD95 for BUSI drops significantly from 5.56 to 4.86. These results demonstrate that multi-view inference during testing can effectively balance prediction noise, mitigate contour jitter, and produce smoother, more consistent outputs.

Building on this, the combination of Geometric Augmentation (rotation and flipping) with TTA proves to be the most effective single-category augmentation strategy in this group. This configuration achieves the best or near-best results across all five datasets. For instance, it attains a Dice of  $87.15 \pm 3.13\%$  on Kvasir-SEG and improves the IoU on GlaS to  $94.30 \pm 0.12\%$ , while significantly reducing the HD95 to 0.99. Furthermore, it continues to deliver performance gains on ETIS, which has large structural variations, and on BUSI, which suffers from significant noise interference. This suggests that this strategy enhances the model's adaptability to pose changes and variations in shape and scale.

Color and Contrast Jittering + TTA primarily improves the model's robustness to illumination differences and local

brightness variations. It leads to noticeable performance gains on endoscopic datasets like Kvasir-SEG and GlaS. However, the improvement is relatively modest on ETIS, where noise constitutes a larger proportion of the challenge, indicating that this augmentation is more suited for scenarios with large brightness and texture variations.

In contrast, Blurring / Elastic Deformation + TTA yields more stable performance on ETIS and GlaS, reducing errors caused by boundary fragmentation (e.g., HD95 decreases to 1.41 on Kvasir-SEG and 0.97 on GlaS). This suggests that such augmentations help the model learn more tolerant structural deformations, thereby improving its ability to model boundaries at different scales.

Finally, Speckle / Gaussian Noise + TTA provides a distinct advantage on BUSI and ETIS (e.g., HD95 on BUSI drops to 3.58, the best result among all configurations for BUSI). This indicates that noise-based augmentations are particularly effective for low-SNR images, as they help suppress misjudgments caused by imaging noise.

**Table 9**

Effects of different data augmentation strategies on GLAS and BUSI datasets (Dice (%), IoU (%), ASSD (mm)). Best results are highlighted in **bold**.

| Augmentation               | GLAS                |                     |                    | BUSI                |                     |                    |
|----------------------------|---------------------|---------------------|--------------------|---------------------|---------------------|--------------------|
|                            | IoU                 | Dice                | HD95↓              | IoU                 | Dice                | HD95↓              |
| None                       | 81.78 ± 0.14        | 89.94 ± 0.08        | 1.78 ± 0.01        | 56.86 ± 3.03        | 67.32 ± 9.75        | 5.56 ± 0.79        |
| None+TTA                   | 82.70 ± 0.35        | 90.53 ± 0.64        | 1.08 ± 0.03        | 62.85 ± 5.88        | 77.07 ± 4.47        | 4.86 ± 1.55        |
| Rotation+Flip+TTA          | 89.13 ± 0.20        | 94.25 ± 0.11        | 0.99 ± 0.03        | 64.99 ± 0.47        | 78.74 ± 0.29        | 3.75 ± 0.53        |
| Color/Contrast Jitter+TTA  | 87.51 ± 0.38        | 93.34 ± 0.23        | 1.04 ± 0.10        | 62.99 ± 1.10        | 77.15 ± 0.97        | 3.68 ± 0.38        |
| Blur/Elastic+TTA           | 87.60 ± 0.56        | 93.39 ± 0.32        | <b>0.97 ± 0.20</b> | 64.58 ± 5.65        | 78.27 ± 4.23        | 3.58 ± 0.48        |
| Speckle/Gaussian Noise+TTA | 88.93 ± 0.88        | 94.13 ± 0.50        | 1.12 ± 0.20        | 66.75 ± 0.97        | 80.03 ± 0.68        | 3.74 ± 0.33        |
| <b>All+TTA</b>             | <b>89.97 ± 0.21</b> | <b>94.30 ± 0.12</b> | 1.09 ± 0.25        | <b>71.30 ± 1.87</b> | <b>83.21 ± 1.26</b> | <b>2.79 ± 0.99</b> |

**Table 10**

Effect of input resolutions on small-object and full-object subsets across three datasets. Higher Dice/IoU indicate better, lower HD95 is better. Best results in **bold**.

| Variant             | Kvasir-SEG          |                     |                    | CVC-ClinicDB        |                     |                    | ETIS                |                     |                    |
|---------------------|---------------------|---------------------|--------------------|---------------------|---------------------|--------------------|---------------------|---------------------|--------------------|
|                     | IoU(%)              | Dice(%)             | HD95↓              | IoU(%)              | Dice(%)             | HD95↓              | IoU(%)              | Dice(%)             | HD95↓              |
| <b>Small (256)</b>  | 87.45 ± 0.91        | 93.54 ± 0.56        | 1.13 ± 0.50        | 89.95 ± 0.13        | 94.70 ± 0.07        | 0.89 ± 0.15        | 89.84 ± 0.19        | 95.08 ± 0.10        | 0.90 ± 0.15        |
| <b>Small (512)</b>  | 87.98 ± 0.86        | 93.84 ± 0.54        | 1.03 ± 0.33        | 90.12 ± 0.05        | 94.79 ± 0.84        | 0.67 ± 0.58        | 90.79 ± 1.88        | 95.15 ± 1.02        | 0.50 ± 0.42        |
| <b>Small (1024)</b> | <b>88.21 ± 4.02</b> | <b>94.11 ± 1.70</b> | <b>0.92 ± 0.57</b> | <b>90.41 ± 1.41</b> | <b>94.80 ± 0.03</b> | <b>0.63 ± 0.52</b> | <b>91.33 ± 1.72</b> | <b>95.46 ± 0.94</b> | <b>0.27 ± 0.46</b> |
| <b>All (256)</b>    | 88.04 ± 0.24        | 93.73 ± 2.07        | 1.05 ± 0.54        | 90.96 ± 0.20        | 95.26 ± 0.10        | 0.31 ± 0.11        | 91.40 ± 0.40        | 95.50 ± 0.22        | <b>0.10 ± 0.14</b> |
| <b>All (512)</b>    | 88.30 ± 3.72        | 94.56 ± 1.35        | 1.00 ± 0.71        | 90.80 ± 0.22        | 95.17 ± 0.12        | 0.39 ± 0.18        | 91.15 ± 0.67        | 95.21 ± 0.02        | 0.18 ± 0.20        |
| <b>All (1024)</b>   | <b>89.04 ± 1.20</b> | <b>94.66 ± 0.45</b> | <b>0.74 ± 0.82</b> | <b>91.03 ± 0.28</b> | <b>95.30 ± 0.15</b> | <b>0.26 ± 0.12</b> | <b>91.74 ± 0.08</b> | <b>95.69 ± 0.04</b> | 0.14 ± 0.20        |

Overall, combining any of the augmentation strategies with TTA consistently improves model performance. Among them, Geometric Augmentation + TTA delivers the most balanced performance across all datasets, making it the comprehensively optimal configuration in this experimental group. Meanwhile, color perturbation, blurring/elastic deformation, and noise augmentation demonstrate targeted strengths in scenarios characterized by large brightness differences, significant structural deformation, or high noise prevalence, respectively.

#### Impact of Resolution on Small Objects Analysis

During training, this study consistently employed the full training set for model learning to ensure comprehensive coverage of targets across different scales and morphologies. To further investigate the effect of input resolution on small structures, we constructed small-object subsets from each dataset based on the SPIE definition of small objects (area proportion < 0.12%), and evaluated performance under three input resolutions during validation: 256×256, 512×512, and 1024×1024. The relevant results are presented in Table 10.

From the validation results, the following observations can be made:

(1) An input resolution of 256×256 did not lead to a degradation in performance on small objects.

On the Kvasir-SEG, CVC-ClinicDB, and ETIS datasets, the Dice and HD95 values at 256 resolution remained comparable to those at higher resolutions, and in some cases were slightly better.

(2) Increasing the resolution to 512 or 1024 did not result in significant overall performance changes.

Differences in metrics among the three resolutions were generally small, typically within a range of ±1%, indicating that the model adapts well to varying input sizes.

(3) The multi-scale architecture of the network helps mitigate the impact of resolution variation.

Thanks to modules such as MFFM, SFR, and xBRD, which model cross-scale semantic and detail information, the model maintains stable performance in both small-object and all-object scenarios.

Overall, changes in resolution had limited impact on the model's ability to recognize small objects. Considering both accuracy and computational cost, 256×256 was selected as the default input scale in this study.

#### Module Replacement Ablation Study

To further validate the balance between performance and complexity of the overall architecture, we conducted a systematic evaluation from three perspectives: parameter count, FLOPs, and segmentation performance across multiple datasets. We designed a module replacement ablation experiment in which RegNet was substituted with ResNet50, EMA was replaced by ECA, and simDB was exchanged with LiteSimDB. The performance was then compared against both the baseline UNet and UNet+CBAM. As shown in Tables 11 and Table 12, the proposed model has 19.005M parameters, representing reductions of 45% and 39% compared to the classic UNet (34.53M) and UNet+CBAM[20] (31.04M), respectively. Its FLOPs are only 4.206G, approximately 1/15 of those of UNet.

Despite the significantly lower computational load, the model achieves the best Dice/IoU performance on the CVC, ETIS, GlaS, and BUSI datasets. Specifically, it attains Dice scores of  $90.79 \pm 0.04\%$  on CVC and  $91.15 \pm 0.67\%$  on ETIS, markedly higher than UNet's  $81.83 \pm 1.23\%$  and  $79.84 \pm 2.34\%$ , respectively.

Of particular note are the changes in the HD95 metric. HD95 reflects boundary deviation and is often more indicative of a medical segmentation model's practical utility than Dice alone. The proposed method reduces HD95 from  $26.45 \pm 3.21$  to  $0.46 \pm 0.20$  on the CVC dataset and from  $35.67 \pm 4.56$  to  $0.22 \pm 0.36$  on ETIS, demonstrating a clear advantage. This indicates that the model possesses stronger robustness in boundary modeling.

In summary, the coordinated design of the proposed MFFM, SFR, and xBlock modules does not incur substantial additional computational overhead. On the contrary, within a lightweight framework, it achieves consistent performance gains across datasets, proving the architecture's effective trade-off between efficiency and accuracy.

#### cross-dataset validation

To comprehensively verify the robustness of the model on heterogeneous datasets, we further constructed a cross-dataset test matrix among Kvasir-SEG, CVC-ClinicDB, and ETIS-LARIB: the model was trained independently on one dataset and then tested on the remaining datasets for inference, thereby evaluating its generalization capability under varying imaging conditions, instrument types, and lesion

**Table 11**

Performance comparison on CVC-ClinicDB and ETIS datasets. Higher Dice/IoU indicate better; lower HD95 is better. Best results are highlighted in **bold**.

| Model       | Param(M)↓ | FLOPs(G)↓ | CVC-ClinicDB      |                   |                  | ETIS              |                   |                  |
|-------------|-----------|-----------|-------------------|-------------------|------------------|-------------------|-------------------|------------------|
|             |           |           | IoU(%)            | Dice(%)           | HD95↓            | IoU(%)            | Dice(%)           | HD95↓            |
| U-Net       | 34.53     | 65.53     | 74.64±1.45        | 81.83±1.23        | 26.45±3.21       | 69.69±2.89        | 79.84±2.34        | 35.67±4.56       |
| U-Net+CBAM  | 31.04     | 54.37     | 76.85±1.32        | 83.52±1.15        | 24.12±3.05       | 71.88±2.65        | 81.45±2.10        | 32.89±4.20       |
| ResNet50    | 35.60     | 7.76      | 85.95±1.15        | 92.44±0.66        | 1.00±0.21        | 87.45±0.44        | 93.79±0.18        | 0.90±0.14        |
| ECA         | 18.90     | 3.61      | 88.58±1.66        | 93.83±0.80        | 2.33±2.24        | 89.44±0.35        | 94.42±0.22        | 1.00±0.00        |
| LiteSIMDB   | 18.09     | 3.69      | 86.95±1.05        | 92.99±0.61        | 1.02±0.34        | 87.72±0.32        | 93.45±0.18        | 0.88±0.12        |
| <b>Ours</b> | 19.01     | 4.21      | <b>90.96±0.20</b> | <b>95.26±0.10</b> | <b>0.31±0.11</b> | <b>91.40±0.40</b> | <b>95.50±0.22</b> | <b>0.10±0.14</b> |

**Table 12**

Performance comparison on GLAS and BUSI datasets. Higher Dice/IoU indicate better; lower HD95 is better. Best results are highlighted in **bold**.

| Model       | GLAS              |                   |                  | BUSI              |                   |                  |
|-------------|-------------------|-------------------|------------------|-------------------|-------------------|------------------|
|             | IoU(%)            | Dice(%)           | HD95↓            | IoU(%)            | Dice(%)           | HD95↓            |
| U-Net       | 87.62±0.29        | 93.35±0.16        | 0.83±0.18        | 71.91±3.54        | 57.22±4.74        | 7.57±2.44        |
| U-Net+CBAM  | 88.95±0.35        | 94.02±0.20        | <b>0.79±0.16</b> | 73.28±3.25        | 59.15±4.35        | 7.15±2.20        |
| ResNet50    | 89.13±1.13        | 93.97±0.25        | 1.19±0.31        | 64.54±6.38        | 78.19±4.49        | 4.61±0.59        |
| ECA         | 88.90±0.43        | 94.16±0.25        | 0.90±0.02        | 61.43±2.50        | 76.05±1.94        | 4.40±0.18        |
| LiteSIMDB   | 87.88±1.10        | 93.55±0.62        | 0.97±0.00        | 61.94±1.79        | 77.15±0.38        | 4.27±0.00        |
| <b>Ours</b> | <b>89.23±0.21</b> | <b>94.30±0.12</b> | 1.09±0.25        | <b>71.30±1.87</b> | <b>83.21±1.26</b> | <b>2.79±0.99</b> |

morphologies. The experimental results are presented in Table 13, where the model demonstrates stable and convincing generalization performance in cross-dataset testing.

Specifically, when trained on Kvasir-SEG and tested on CVC-ClinicDB and ETIS-LARIB, the model achieved IoU scores of  $79.27 \pm 1.33$  and  $75.95 \pm 0.76$ , respectively, while maintaining an HD95 of  $14.66 \pm 2.25$  on ETIS. When CVC-ClinicDB was used as the training set, the model maintained reasonable performance even under more pronounced domain shifts, obtaining IoU scores of  $64.37 \pm 0.06$  on Kvasir-SEG and  $65.28 \pm 3.41$  on ETIS-LARIB. Training on ETIS-LARIB led to even more pronounced stability on external distributions, achieving an IoU of  $79.98 \pm 0.16$  on CVC-ClinicDB and retaining a low boundary error (HD95:  $4.25 \pm 0.11$ ) on Kvasir-SEG.

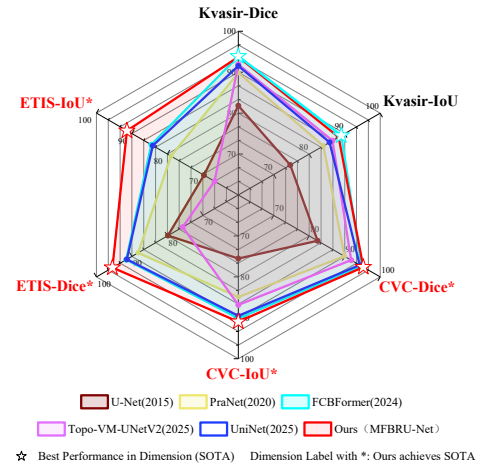
These results indicate that, even under strict patient-wise data partitioning, the model can maintain reliable performance. The cross-dataset evaluation further confirms its adaptability to domain variations introduced by different imaging sources, resolutions, and device conditions, demonstrating robust generalization capability in scenarios with limited samples, noise interference, and significant external distribution shifts.

Overall, by introducing patient-wise partitioning and systematic cross-dataset validation, this study significantly enhances the rigor of the experimental design, making the evaluation results more objective and credible.

## 5. Conclusion

This paper addresses key challenges in medical image segmentation—such as structural ambiguity, semantic

inconsistency, and boundary degradation—by proposing a novel segmentation network named MFBRU-Net. Figure 8 (adjusted for this section) illustrates the overall architecture and information flow, facilitating an understanding of the interaction mechanisms between its modules. MFBRU-Net constructs a comprehensive information enhancement pathway through three core components:



**Figure 8:** Performance overview of MFBRU-Net against state-of-the-art medical image segmentation methods .

(1) The MFFM enables efficient aggregation of shallow detail and deep semantic features via a unified scale and channel design.

**Table 13**

Cross-dataset evaluation. Rows indicate the training dataset and columns indicate the testing dataset.

| Train  | Kvasir       |              |             | CVC-ClinicDB |              |              | ETIS         |              |              |
|--------|--------------|--------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|
|        | IoU (%) ↑    | Dice (%) ↑   | HD95 ↓      | IoU (%) ↑    | Dice (%) ↑   | HD95 ↓       | IoU (%) ↑    | Dice (%) ↑   | HD95 ↓       |
| Kvasir | —            | —            | —           | 66.50 ± 1.83 | 79.27 ± 1.33 | 6.81 ± 1.16  | 62.11 ± 0.94 | 75.95 ± 0.76 | 14.66 ± 2.25 |
| CVC    | 64.37 ± 0.06 | 78.10 ± 0.07 | 5.45 ± 0.17 | —            | —            | —            | 49.38 ± 3.62 | 65.28 ± 3.41 | 19.69 ± 3.26 |
| ETIS   | 67.12 ± 0.57 | 79.98 ± 0.16 | 4.25 ± 0.11 | 55.00 ± 1.49 | 70.29 ± 1.41 | 10.93 ± 0.37 | —            | —            | —            |

(2) The SFR, incorporating EMA attention and a simDB semantic distillation strategy, promotes synergistic alignment of multi-scale features.

(3) The xBRD integrates an enhanced xLSTM structure within the decoding path, significantly strengthening contextual relationship modeling while effectively mitigating boundary degradation.

Experimental results on five public medical image segmentation datasets (GlaS, BUSI, ETIS, CVC-ClinicDB, Kvasir-SEG) demonstrate that MFBRU-Net outperforms mainstream methods in key metrics such as Dice and IoU. Its performance is particularly outstanding in scenarios involving high structural complexity, blurred lesion boundaries, or irregular tissue morphology. Furthermore, ablation studies confirm the independent contributions and synergistic benefits of each module, validating the rationality and effectiveness of the design strategy.

#### Discussion of Limitations

Although the experimental results demonstrate a degree of generalizability and stability, this study has several limitations. First, while the five public datasets are clinically sourced, they have undergone certain levels of screening, cropping, and preprocessing. Their complexity may not fully represent real clinical environments, which involve factors such as imaging noise, device heterogeneity, occlusions, and extreme morphological variations. Consequently, the model's robustness in real-world, multi-center, and multi-device settings requires further evaluation. Second, this work primarily focuses on 2D image segmentation tasks and does not address more challenging scenarios such as 3D imaging, temporal structures, or cross-modal joint segmentation. Additionally, although the current model exhibits lightweight characteristics, there remains potential for further compression and acceleration for deployment on resource-constrained devices (e.g., portable ultrasound systems or intraoperative navigation systems).

In future work, we plan to extend our research in the following directions:

(1) Extending MFBRU-Net to 3D medical imaging and cross-modal segmentation tasks, exploring its adaptability in multi-modal contexts involving MRI, CT, and ultrasound.

(2) Incorporating few-shot learning, weak supervision, and semi-supervised methods to enhance the model's applicability in real clinical settings with scarce annotations.

(3) Further investigating the model's interpretability mechanisms—such as visualizing attention weights and semantic flow maps—to improve clinical acceptance.

(4) Developing more efficient ultra-lightweight inference frameworks tailored for practical deployment, supporting critical scenarios like real-time endoscopic recognition and intraoperative navigation.

(5) Collaborating closely with clinical experts to promote the practical application of MFBRU-Net in tasks such as tumor boundary annotation, polyp detection, and tissue analysis.

In conclusion, the modular design proposed in MFBRU-Net offers a new perspective for addressing semantic inconsistency and boundary degradation in medical image segmentation. As multi-modal and multi-center data, along with more diverse weakly-supervised learning strategies, are incorporated in the future, the practical clinical potential of this approach warrants further in-depth exploration.

#### CRedit authorship contribution statement

**Han Gao:** Conceptualization of this study, Methodology, Software, Writing - Original Draft, Investigation, Data Curation, Visualization . **Xianran Zhang:** Resources . **Zhengpeng Li:** Funding acquisition . **Ziteng Wang:** Visualization . **Ping Sun:** Visualization . **Jian-sheng Wu:** Writing – Review & Editing , Supervision .

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Ethics declarations

##### Ethical approval

This study used only publicly available and fully anonymized datasets. Therefore, ethical approval was not required, as no human or animal subjects were directly involved.

##### Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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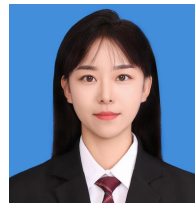
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